

**CARGOMATE**  
**SMART ROBOTIC INVENTORY SYSTEM WITH CLOUD**  
**INTEGRATION**

**25J-26J-032**

Design Document (DD) of Autonomous Navigation System

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# 1. INTRODUCTION

## 1.1 Purpose

The purpose of this Design Document (DD) is to provide a comprehensive and detailed description of the architectural design, component interactions, and implementation strategies of the cloud integration developed as part of this project. This document serves as a technical reference for academic evaluators, supervisors, and future developers by explaining not only what components are used, but also why specific design decisions were made and how they collectively satisfy system requirements related to reliability, scalability, security, and maintainability.

Unlike a Software Requirements Specification (SRS), which focuses primarily on what a system should do, this Design Document emphasizes how the system is constructed. Special attention is given to the integration of embedded ESP32 devices, a web-based dashboard with Role-Based Access Control (RBAC), a digital twin for spatial visualization, and a Grafana-based monitoring and alerting infrastructure.

## 1.2 Scope

The scope of this document is limited to the components that were designed and implemented during the current development phase of the Smart Warehouse project. The document covers both software and embedded system aspects, focusing on the following key areas:

- Design and implementation of a secure dashboard with RBAC
- Digital twin modeling of warehouse assets and operational states
- ESP32-based health monitoring and telemetry reporting (in progress)
- System observability using Prometheus and Grafana
- Cloud database integration for persistent storage

Out-of-scope elements include advanced predictive analytics, autonomous control logic, and large-scale industrial deployment considerations, which are identified as future work.

### 1.3 Definitions, Acronyms, and Abbreviations

| Term         | Description                                                                            |
|--------------|----------------------------------------------------------------------------------------|
| RBAC         | Role-Based Access Control mechanism used to restrict system access based on user roles |
| Digital Twin | A virtual representation of physical warehouse assets synchronized with real-time data |
| ESP32        | Wi-Fi enabled microcontroller used for device health monitoring and telemetry          |
| Grafana      | Open-source platform for monitoring, visualization, and alerting                       |
| Cloud DB     | Cloud-hosted database service used for centralized data persistence                    |
| DD           | Design Document                                                                        |

### 1.4 Overview

The Smart Warehouse system integrates operational visualization, infrastructure monitoring, and embedded device health checks into a single unified platform. The dashboard provides real-time insight into warehouse operations, while the digital twin offers an intuitive spatial understanding of asset states. In parallel, Grafana continuously monitors the availability and performance of system components, including ESP32 devices, ensuring that failures are detected early and communicated through automated alerts.

By combining these elements, the system moves beyond traditional static dashboards and enables proactive system management, improved operational awareness, and increased reliability in warehouse environments.

## 2. OVERALL DESCRIPTION

### 2.1 Product Perspective

Conventional warehouse monitoring solutions often focus exclusively on business metrics such as inventory counts and order statuses, while neglecting the health of the underlying systems and devices responsible for data generation. As a result, failures at the device or infrastructure level may go unnoticed until they directly impact operations.

The proposed Smart Warehouse system addresses this limitation by integrating embedded ESP32 devices, cloud services, and monitoring tools into a cohesive architecture. This approach ensures that operational data, system health metrics, and infrastructure status are all visible through a single platform, significantly improving transparency and fault diagnosis capabilities.

#### 2.1.1 System Interfaces

The system interfaces consist of several interacting components. The web dashboard communicates with backend services to retrieve operational data and enforce RBAC policies. ESP32 devices interface with the backend over a wireless network to transmit health information.

#### 2.1.2 User Interfaces

Multiple user interfaces are provided to accommodate different user roles. The login interface authenticates users and applies RBAC policies. Administrators access an administrative dashboard for user management and monitoring oversight, while standard users access a restricted dashboard displaying only permitted operational views. Additionally, a dedicated digital twin interface visualizes warehouse assets in a spatially meaningful manner.

#### 2.1.3 Hardware Interfaces

The primary hardware interface in the system is the ESP32 microcontroller. ESP32 device connects to the network via Wi-Fi and may interface with sensors depending on deployment requirements. The ESP32 acts as a lightweight edge device responsible for reporting its operational health to the backend system.

#### 2.1.4 Software Interfaces

Software interfaces include the backend application programming interfaces (APIs) used by the dashboard and digital twin and the cloud database APIs for persistent storage.

#### 2.1.5 Communication Interfaces

All user-facing communication is conducted over secure HTTPS connections to protect authentication credentials and sensitive data. Internal communication between ESP32 devices and the backend may use MQTT, depending on configuration. Alert notifications are delivered via email or webhook-based integrations.

#### 2.1.6 Memory Constraints

ESP32 devices operate under strict memory constraints, including limited RAM and flash storage. To accommodate these limitations, the firmware is designed to transmit minimal health payloads at fixed intervals.

#### 2.1.7 Operations

During normal operation, users authenticate and interact with the dashboard. In parallel, ESP32 devices periodically transmit heartbeat messages. The backend processes these messages, updates the cloud database, and exposes aggregated metrics. Grafana evaluates alert conditions based on predefined thresholds.

#### 2.1.8 Site Adaptation Requirements

Deployment requires stable network connectivity, proper configuration of ESP32 devices, and valid cloud database credentials. Monitoring thresholds and alert policies must be adapted to the specific scale and operational characteristics of the deployment site.

## 2.3 User Characteristics

Admin users have technical knowledge and manage users and monitoring. Warehouse staff use the dashboard for operational awareness. IT/Operators respond to alerts and system issues.

## 2.4 Constraints

System performance depends on network availability, ESP32 hardware limitations, and cloud service reliability.

## 2.5 Assumptions and Dependencies

It is assumed that ESP32 devices have stable Wi-Fi connectivity and that cloud services are accessible during system operation.

# 3. SPECIFIC REQUIREMENTS (DESIGN DOCUMENT)

## 3.1 External Interface Requirements

The system must provide clearly defined interfaces between users, embedded devices, and monitoring components. These interfaces ensure modularity and allow individual components to evolve without disrupting the overall system.

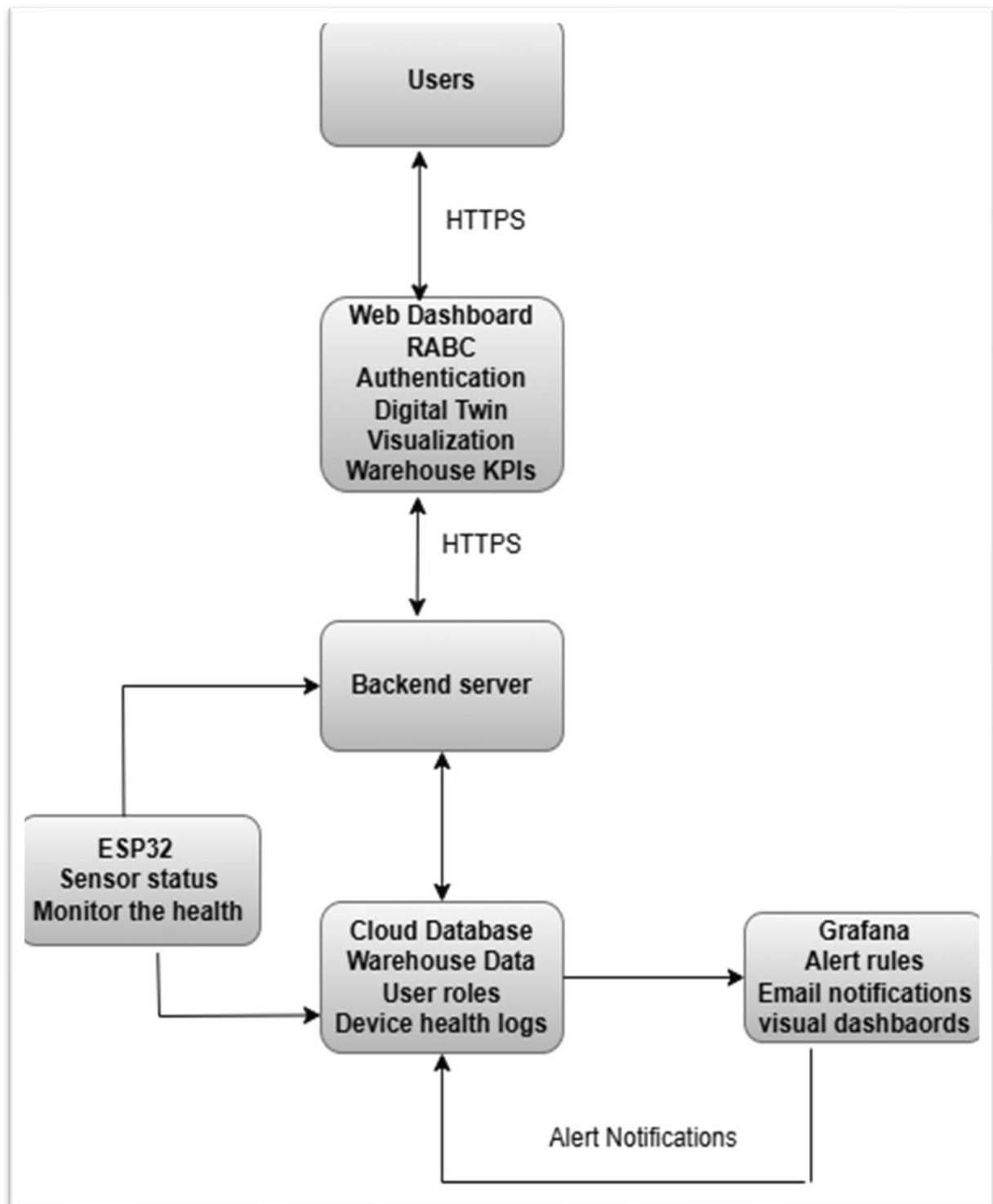
## 3.2 Architectural Design

The architecture follows a layered design approach consisting of an embedded device layer, a backend service layer, a monitoring and observability layer, and a presentation layer. This separation of concerns simplifies maintenance and enhances system scalability.

### 3.2.1 High-Level Architecture Design

At the lowest level, ESP32 devices act as edge nodes responsible for monitoring their own operational status. These devices transmit heartbeat signals to the backend, which processes and aggregates health information. Grafana visualizes these metrics and triggers alerts when abnormal conditions are detected.





### 3.2.2 Hardware & Software Requirements with Justification

The ESP32 microcontroller was selected due to its low cost, integrated Wi-Fi capability, and sufficient processing power for health monitoring tasks. Grafana was chosen for monitoring due to their open-source nature, flexibility, and wide adoption in industry. A cloud database provides durability and scalability for operational data.

### 3.2.3 Risk Mitigation Plan with Alternative Solution Identification

Potential risks include network outages, false alerts, and device failures. Network outages are mitigated through timeout-based health detection. Alert fatigue is reduced by carefully tuning thresholds. In the event of monitoring failure, alternative notification channels may be introduced.

| Risk                                        | Potential Impact                                  | Mitigation Strategy                                              | Alternative Solution                               |
|---------------------------------------------|---------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------|
| Dashboard server downtime                   | Users unable to access dashboard and digital twin | Continuous health monitoring using Grafana with alerting         | use cloud-managed hosting                          |
| RBAC misconfiguration                       | Unauthorized access to sensitive system features  | Role-based access enforcement and access validation during login | Manual admin review and role reassignment          |
| Grafana alert failure (email not sent)      | Administrators not notified during system failure | Test alerts and periodic alert rule validation                   | Webhook-based alerts (SMS / messaging platform)    |
| Cloud database service outage               | Inability to store or retrieve operational data   | Use managed cloud DB with high availability                      | Switch to alternative cloud region                 |
| Incorrect digital twin state representation | Misleading operational information                | Periodic data refresh and validation checks                      | Manual state override or visual warning indicators |
| Data inconsistency between systems          | Incorrect monitoring or reporting                 | Centralized cloud database as single source of truth             | Scheduled data reconciliation process              |

### 3.2.4 Cost–Benefit Analysis

The system leverages open-source tools and low-cost hardware, minimizing financial cost while delivering significant benefits such as improved visibility, reduced downtime, and enhanced operational reliability.

## 3.3 Performance Requirements

The system is designed to detect failures within the configured monitoring interval. Dashboard response times must remain acceptable to support real-time decision-making.

## 3.4 Design Constraints

Hardware limitations of ESP32 devices and network latency impose constraints on update frequency and data payload size.

## 3.5 Software System Attributes

### 3.5.1 Reliability

Reliability is enhanced through continuous monitoring, periodic health checks, and proactive alerting mechanisms.

### 3.5.2 Availability

The separation of monitoring infrastructure from application logic ensures that system availability can be assessed even during partial failures.

### 3.5.3 Security

Security is achieved through RBAC enforcement, secure communication channels, and controlled access to monitoring interfaces.

### 3.5.4 Maintainability

A modular architecture and clear interface definitions facilitate long-term maintainability and future expansion.

## 4. REFERENCES

### References

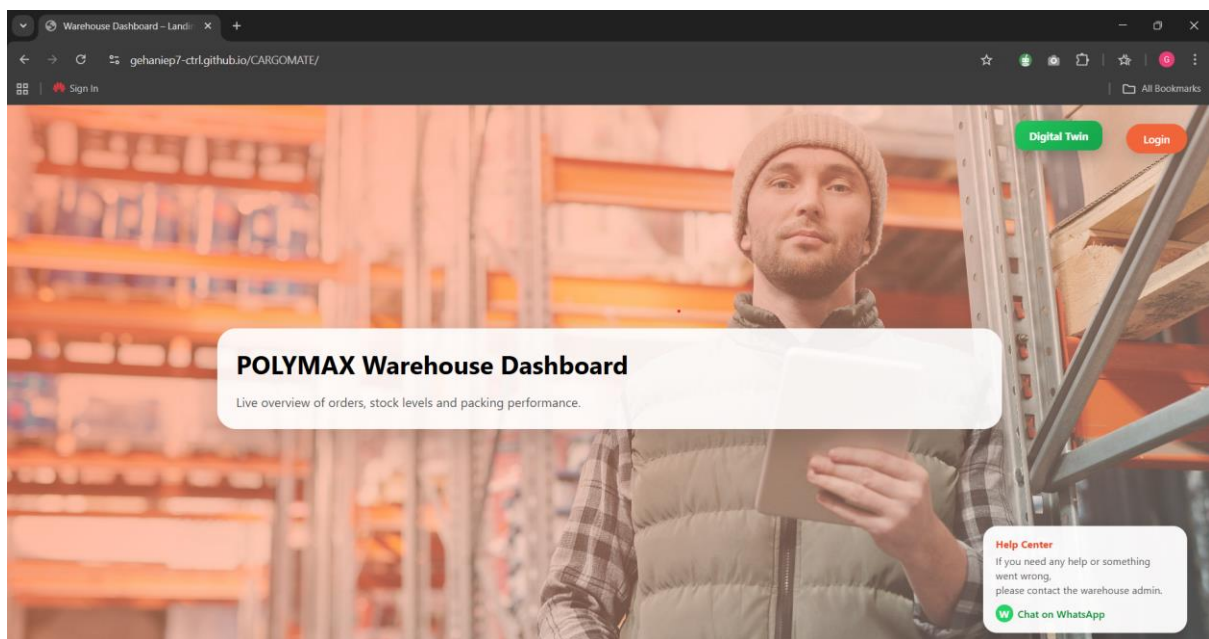
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## 5. APPENDICES

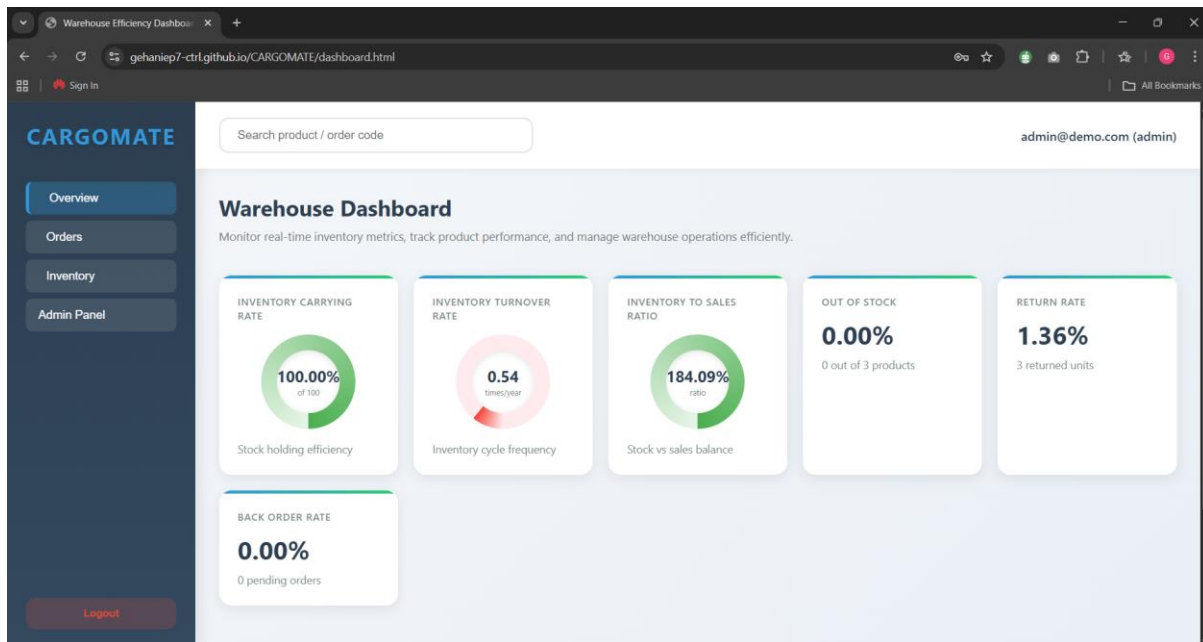
### Appendix A: Development Progress and Implementation Evidence

#### Appendix A.1 – Dashboard Development Progress

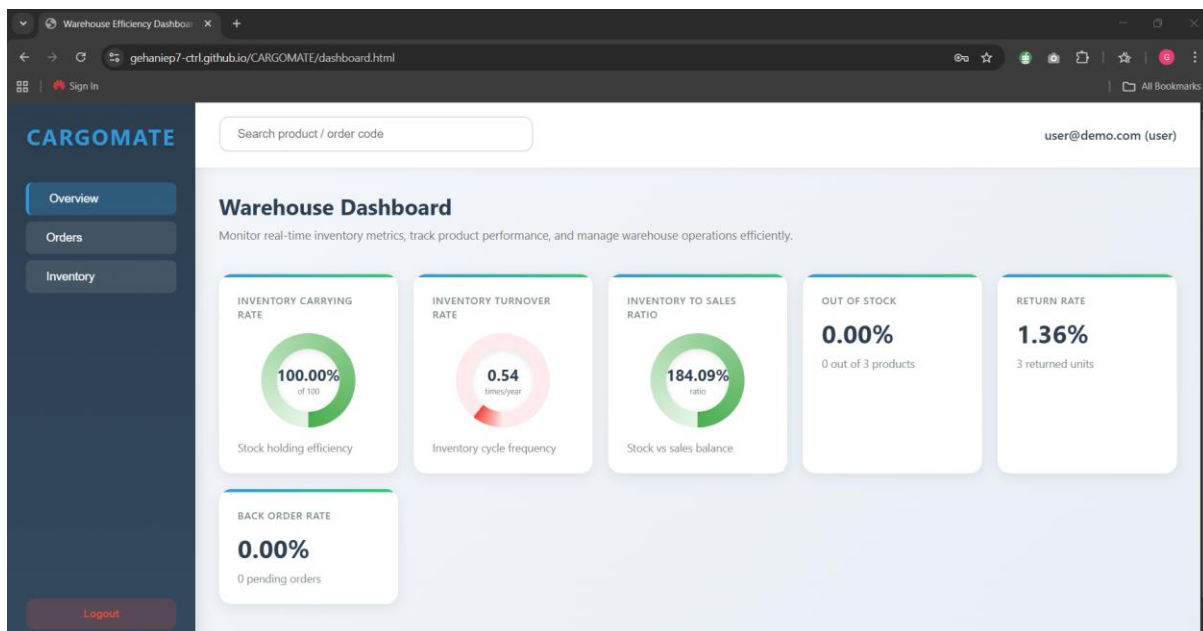
Login page - Login interface implementing role-based access control



## Admin dashboard - Admin dashboard displaying system management features

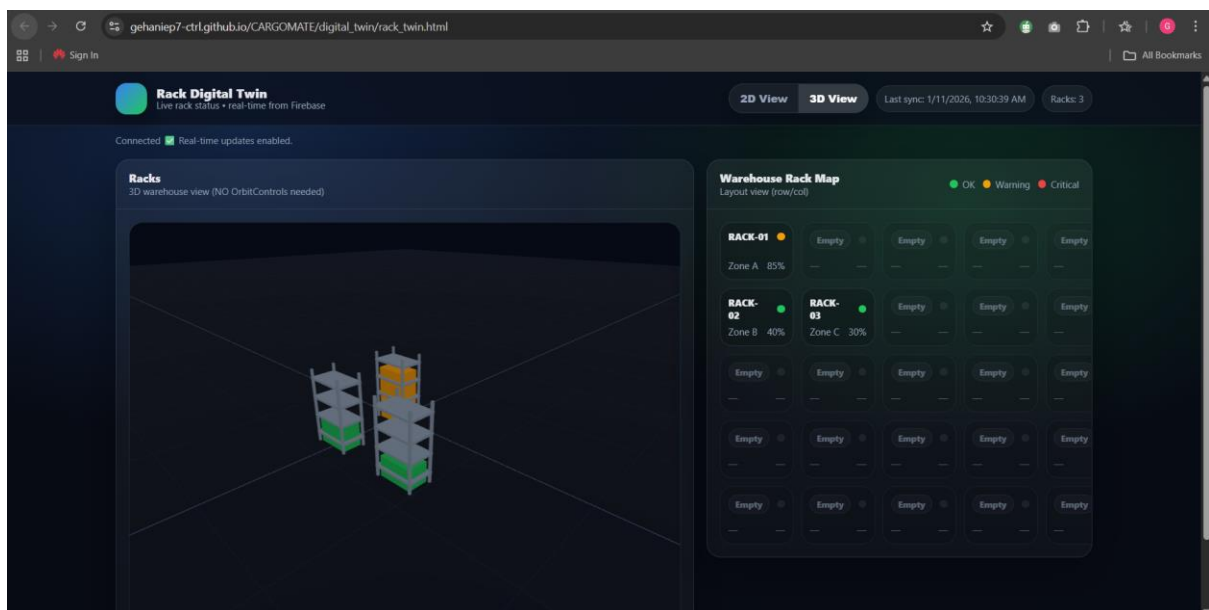
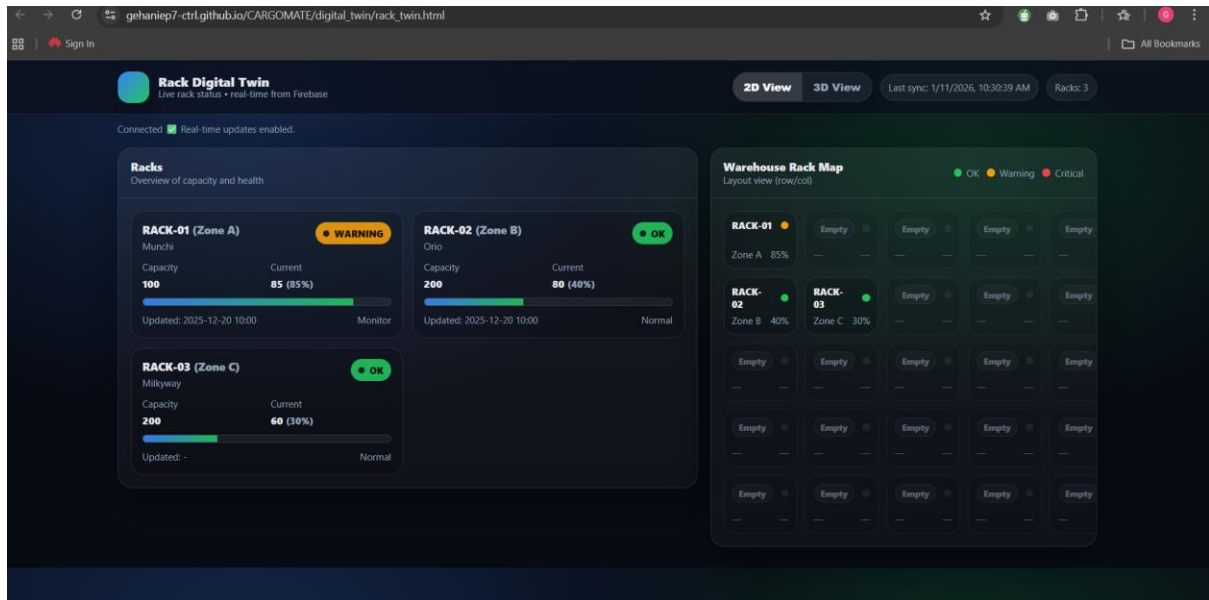


## User dashboard - User dashboard with restricted access based on RBAC



## Appendix A.2 – Digital Twin Implementation Progress

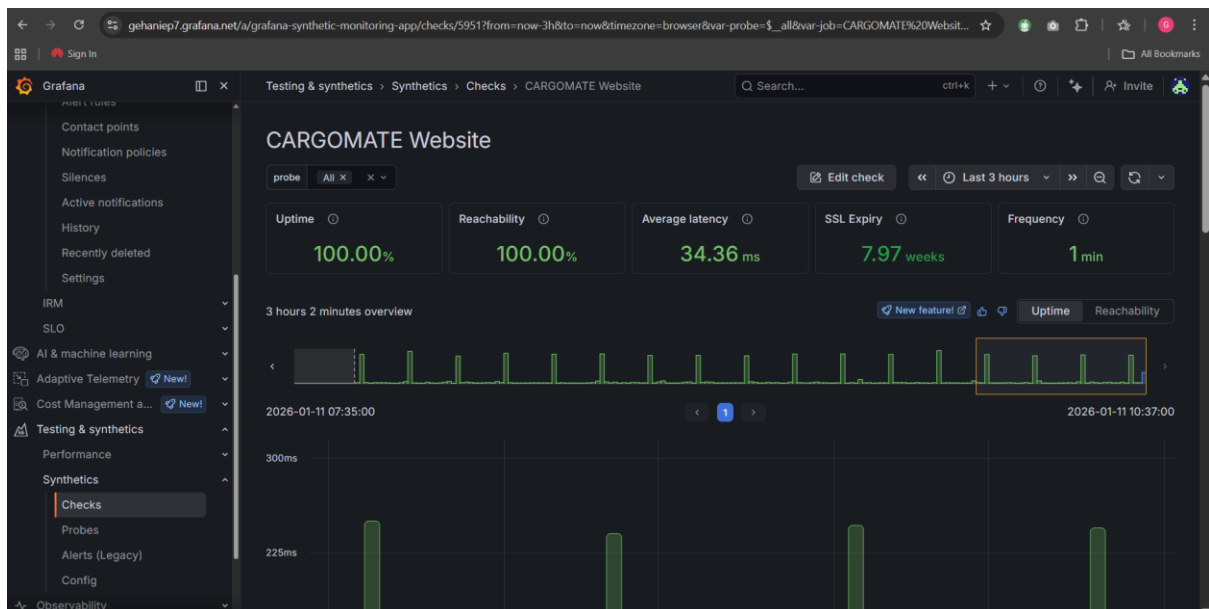
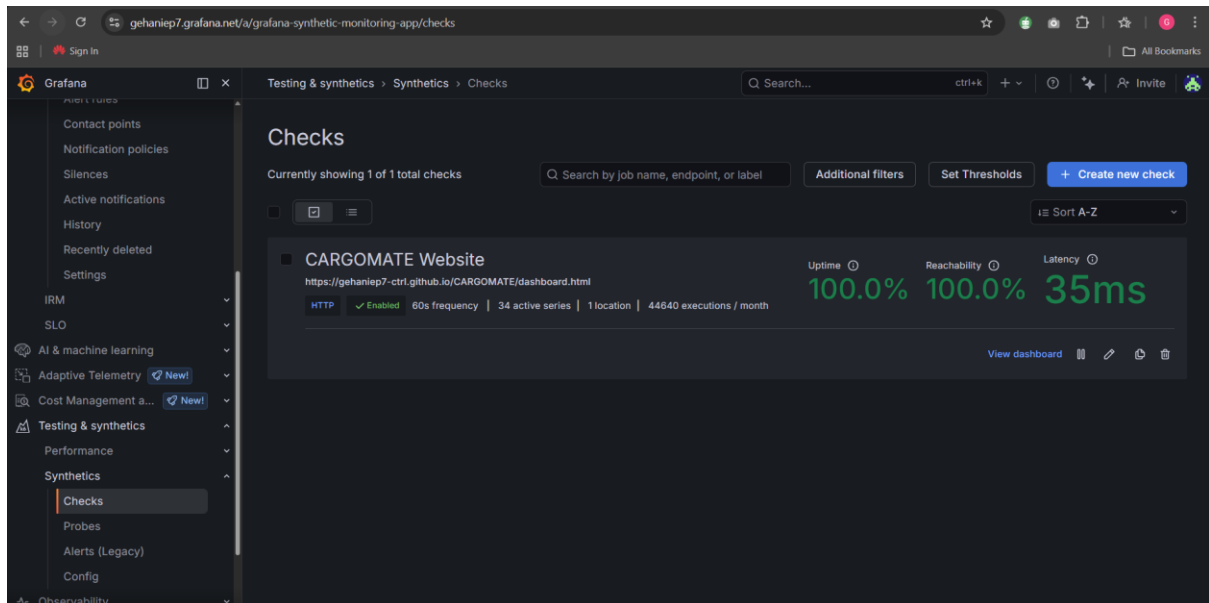
The digital twin provides a virtual representation of the physical warehouse, enabling intuitive spatial monitoring of asset states in real time.



## Appendix A.3 – Grafana Monitoring Dashboard

Grafana is used to continuously monitor system health and availability, ensuring early detection of service failures.

### Grafana dashboard





## Appendix A.6 – Cloud Database Integration

Sample database created in the firebase.

